الأكاديمية العربية للعلوم والتكنولوجيا والنقل البحري

Arab Academy for Science, Technology & Maritime Transport

12th Week Project

Introduction to programming

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1. Student Management System

This system manages student academic records.

Functionality:

- Add new students with name and ID
- Store marks for multiple subjects
- Calculate total and average marks
- Assign grades based on performance
- Display all student records
- Search for a student by name or ID
- Update student information if needed

2. Banking System

A simulation of basic banking operations.

Functionality:

- Create a new bank account with user details
- Assign an initial balance
- Deposit money into the account
- Withdraw money with balance validation
- Display current balance
- Maintain a transaction history
- Prevent overdrawing from account

3. Library Management System

Manages book borrowing and return processes.

Functionality:

- Add books with title, author, and ID
- Display available and borrowed books
- Borrow a book if available
- Return a borrowed book
- Search for a book by title or author
- Track book availability status

4. Inventory Management System

Tracks product stock in a store.

Functionality:

- Add new products with price and quantity
- Update product quantity
- Remove products from inventory
- Display all products
- Calculate total inventory value
- Alert when stock is low

5. Expense Tracker

Tracks daily and monthly expenses.

Functionality:

- Add expenses with category and amount
- Store expense history
- Display all recorded expenses
- Calculate total expenses
- Filter expenses by category
- Generate monthly spending report

6. Quiz Application

A multiple-choice quiz system.

Functionality:

- Load questions and options
- Display one question at a time
- Accept user answers
- Check correctness of answers
- Keep track of score
- Display final result with performance

7. To-Do List Manager

Manages daily tasks efficiently.

Functionality:

- Add new tasks
- Edit existing tasks
- Delete tasks
- Mark tasks as completed

- Display all tasks with status
- Store tasks for future use

8. Attendance System

Tracks student attendance records.

Functionality:

- Mark students as present or absent
- Store attendance for each day
- Display attendance records
- Calculate attendance percentage
- Generate student attendance report

9. Login & Registration System

Handles user authentication.

Functionality:

- Register new users with username and password
- Validate password strength
- Allow user login
- Limit login attempts for security
- Check if username already exists
- Display login success or failure messages

10. Password Strength Checker

Evaluates password security.

Functionality:

- Check password length
- Verify presence of numbers
- Check for uppercase letters
- Check for special characters
- Classify password as weak, medium, or strong
- Provide feedback for improvement

11. Text Analyzer Tool

Analyzes input text data.

Functionality:

- Count total words
- Count total characters
- Count sentences
- Identify most frequently used word
- Convert text to lowercase/uppercase
- Display summary report

12. ATM Simulation System

Simulates ATM operations.

Functionality:

- User login using PIN
- Check account balance
- Deposit money

- Withdraw money with validation
- Show transaction history
- Prevent incorrect PIN access

13. Number System Converter

Converts between number systems.

Functionality:

- Convert decimal to binary
- Convert binary to decimal
- Convert decimal to hexadecimal
- Convert hexadecimal to decimal
- Validate input values
- Display conversion results clearly

14. Sorting & Searching Tool

Performs basic data operations.

Functionality:

- Accept list of numbers from user
- Sort numbers in ascending order
- Search for a specific number
- Display sorted list
- Show search results (found/not found)

15. Multiplication Table Generator

Generates multiplication tables.

Functionality:

- Generate table for a single number
- Generate tables for a range of numbers
- Display formatted output
- Validate user input
- Allow repeated generation

16. Calendar Generator

Displays monthly calendar.

Functionality:

- Input month and year
- Calculate correct starting day
- Display formatted calendar layout
- Show number of days in month
- Handle leap years correctly

17. File-Based Chat Simulation

Simulates messaging system using files.

Functionality:

- Send messages from user A to user B
- Store messages in a file
- Read and display chat history

- Clear chat history if needed
- Simulate conversation flow

18. Mini CRM System

Manages customer leads and follow-ups.

Functionality:

- Add new customer leads
- Store contact details
- Assign lead status (hot/warm/cold)
- Update lead status
- Search for leads
- Generate follow-up reports

19. Grade Calculator System

Calculates academic performance.

Functionality:

- Input marks for multiple subjects
- Calculate total and average
- Assign grades based on range
- Display GPA or percentage
- Show final result report

20. Text-Based Adventure Game

Interactive story game with choices.

Functionality:

- Start game with introduction story
- Present choices to user
- Change story based on decisions
- Manage player inventory
- Handle multiple endings
- Restart or exit game option